



Alternating Optimization for LLM Post-Training: From Divergence to Verified Solutions

The Chinese University of Hong Kong — Department of Computer Science and Engineering

jianghuanyun

Supervisor: Prof. Hoi To Wai

ALS & its 3 repairs → DIVERGE | SGD / AdamW / LoRA → CONVERGE | proof below (matched budgets, pretrained baselines shown)

Research Question

Why does ALS-based post-training diverge on deep LLMs, and what works instead? ALS solves $W^* = (X^T X + \lambda I)^{-1} X^T Y$ in closed form — one solve replacing hundreds of gradient steps. It converges on ≤ 24 -layer models but diverges on every observed 28-layer attempt. Two sub-questions: (Q1) can the divergence be explained and fixed? (Q2) what actually works, compared fairly?

The 2×2 Framework — Four Post-Training Algorithms

Two design choices define how a model is fine-tuned: **which optimizer** drives the update, and **how many parameters** are trained. Crossing them gives four algorithms. Here is each one, in plain terms:

	Full-rank train ALL weights	LoRA train a small low-rank adapter
AdamW adaptive-step gradient descent + weight decay	B: repeatedly compute each weight's slope (gradient) of the loss and step downhill; AdamW gives each weight an $(r \times d)$ and B $(d \times r)$ whose product $\Delta W = BA$ ($r \ll d$) auto-tuned step size, so it descends quickly and stably. Updates every weight of the model. 1.25 PPL.	D: freeze the whole model; train only two small matrices A and B (d×r) whose product $\Delta W = BA$ (r×d) approximates a full update with far fewer parameters. 10.41 PPL.
ASP alternating: ALS closed-form solve + SGD + noise	A: each cycle, ALS computes the least-squares-optimal output weights $W^* = (X^T X + \lambda I)^{-1} X^T Y$ in one matrix solve — bypassing thousands of gradient steps — then SGD and noise follow. DIVERGES on deep models.	C: the same ALS + gradient + noise cycle, applied to the small adapter instead of full weights. Converges, but quality is an order of magnitude worse than AdamW. 135.4 PPL.

All four run until they consume the **same total FLOPs** (one ALS solve $\gg$ one AdamW step), with shared data, seeds, and evaluation — so the comparison is fair.

The Story in 4 Steps

STEP 1 — ALS diverges on deep models

The closed-form solve $W^* = (X^T X + \lambda I)^{-1} X^T Y$ converges on ≤ 24 -layer models but diverges on every observed 28-layer attempt (11/11 on Qwen2.5-7B; PPL → NaN in 1–3 cycles).

STEP 2 — Mechanism proven by controlled ablation

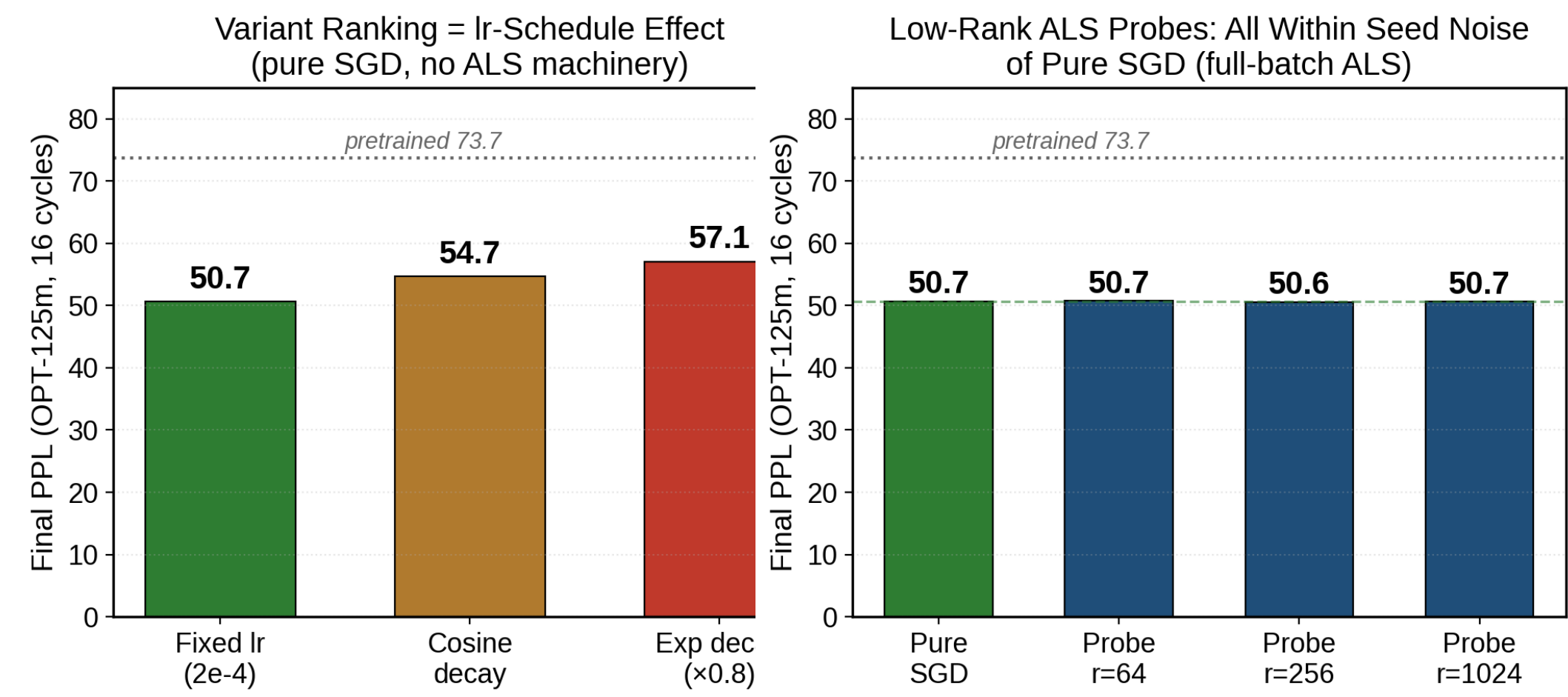
Five-condition ablation on Qwen2.5-7B isolates **ALS weight modification** as the necessary-and-sufficient cause within the tested space: every condition with a real weight solve diverges; SGD-only, perturbation-only and suppressed-weight ALS all converge (Fig. 5).

STEP 3 — All three natural repairs fail matched controls

Gradient injection, low-rank probing and soft-target distillation — each matched against an equal-budget SGD baseline — add no measurable value (Figs. 1–3).

Repair analysis — Variant ranking = lr-schedule effect; probes within noise

With injection vacuous, the ranking is the lr schedule alone (fixed 50.7 < cosine 54.7 < exp 57.1). Probe ranks $r=64/256/1024$ all sit within seed noise of pure SGD (50.7/50.6/50.7 vs 50.7) — the closed-form solve is redundant with the CE gradient.



STEP 4 — Verified alternatives, baselines shown

AdamW full-rank 1.25 PPL; AdamW+LoRA 10.41 PPL at 0.1% trainable params; plain SGD 6.83 PPL. All far below the pretrained 73.1 PPL reference — **up to 58× improvement over doing nothing** (Fig. 4).

Contributions

- Diagnosis** — first controlled evidence that ALS weight modification (not noise / optimizer / hook overhead) causes deep-model divergence, via a 5-condition ablation.
- Theory** — residual-amplification model ($p \approx 1.08/\text{layer}$) predicts a 25–28-layer boundary, consistent with 8 architectures.
- Saved design space** — three plausible ALS repairs proven ineffective by matched-budget controls; identifies why (closed-form solve $\equiv$ CE gradient).
- Verified alternatives** — AdamW 58×, SGD 10.7×, LoRA 7× below the 73.1 pretrained baseline, with a corrected evaluation harness.

Method & discipline

- 2×2 factorial** (ASP vs AdamW × full-rank vs LoRA), FLOPs-matched, shared seeds/data.
- Harness audit:** OPT numbers re-verified with correct tokenizer + pad-masked labels (baseline 73.7 PPL); 7B baseline 73.1 PPL.
- Falsification discipline:** pre-stated hypothesis, open code, negative results reported with matched controls.

Verified Solutions (Qwen2.5-7B, 28 layers)

AdamW full-rank — 1.25 PPL (58× below baseline). Algorithm core: for every weight, compute the loss gradient (the slope that tells which way is downhill), then step down with an **adaptive per-weight step size** (Adam's moment estimates) plus weight decay. Repeated thousands of times, it descends the loss surface. It converges reliably at any depth and achieves the best absolute quality (N=3, 800 steps). *Use when:* quality is the priority and training compute is available.

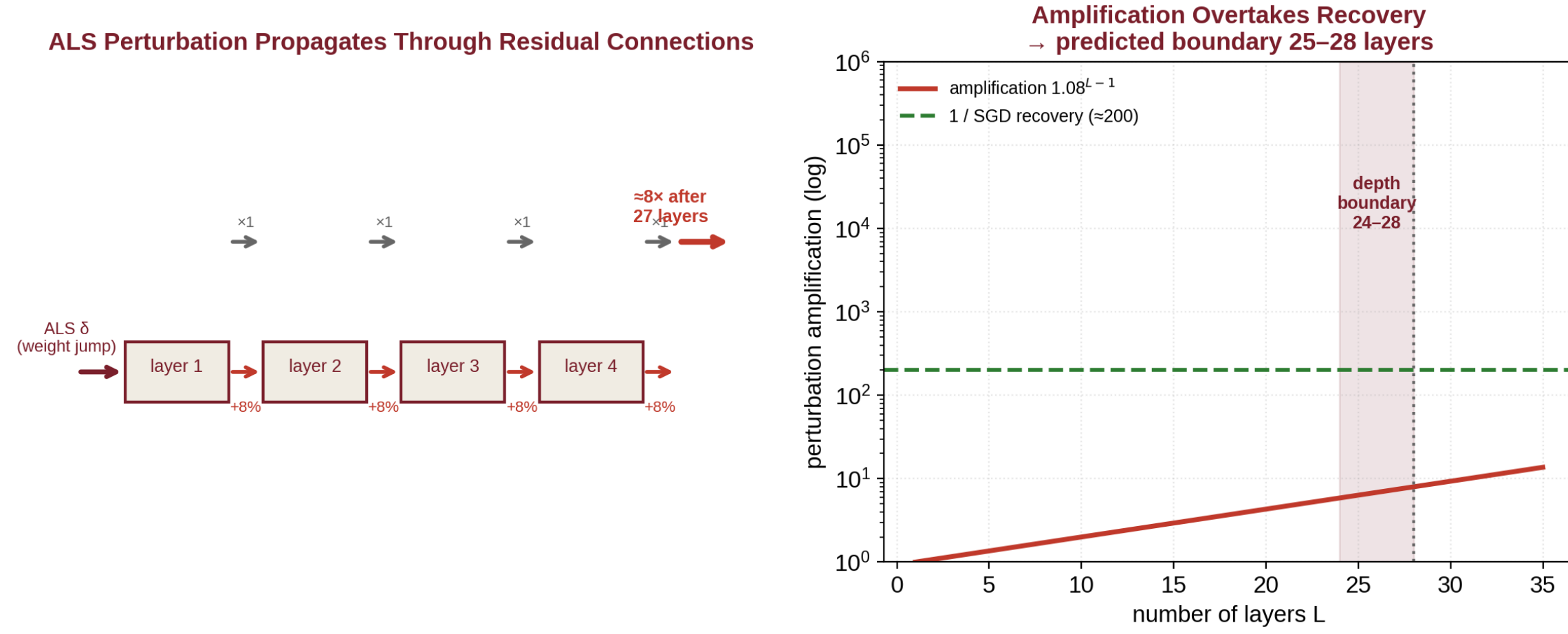
AdamW + LoRA — 10.41 PPL (7× below baseline). Algorithm core: freeze the entire base model and train only a **tiny adapter** $\Delta W = BA$ (A: $r \times d$, B: $d \times r$, $r \ll d$, $\sim 0.1\%$ of parameters) with the same AdamW updates. The adapter's low rank forces a compact, generalizable correction. Nearly the same FLOPs as full-rank (backprop through frozen weights dominates), but the optimizer state and adapter memory are orders of magnitude smaller. *Use when:* storage/memory is constrained or many task-adapters must be served.

Plain SGD, fixed lr — 6.83 PPL (10.7× below baseline). Algorithm core: the simplest descent — for every weight, take a small step opposite the gradient with a **single fixed step size (learning rate)**, never adapting it per-weight and never decaying it over training. It converges on 28 layers where ALS cannot — the decisive lesson that the failure lives in the ALS solve, not in gradient training. *Use when:* a simple, stable, dependency-free baseline is needed.

Method	Final PPL	vs 73.1 baseline	Best for
AdamW full-rank	1.25	58× better	absolute quality
Plain SGD (fixed lr)	6.83	10.7× better	simple stable baseline
AdamW + LoRA	10.41	7× better	memory / many adapters
ASP full-rank	DIVERGED	11/11 attempts	—

Why ALS Diverges — The Theory

Residual connections amplify an ALS perturbation by an **effective $p \approx 1.08$ per layer**: the identity path preserves it exactly, and each layer's nonlinearity adds $\sim 8\%$. Over 27 layers that is $\approx 8\times$ **amplification** — beyond SGD's per-cycle recovery (≈ 0.005), a 1600:1 asymmetry: the model cannot heal faster than the perturbation grows.

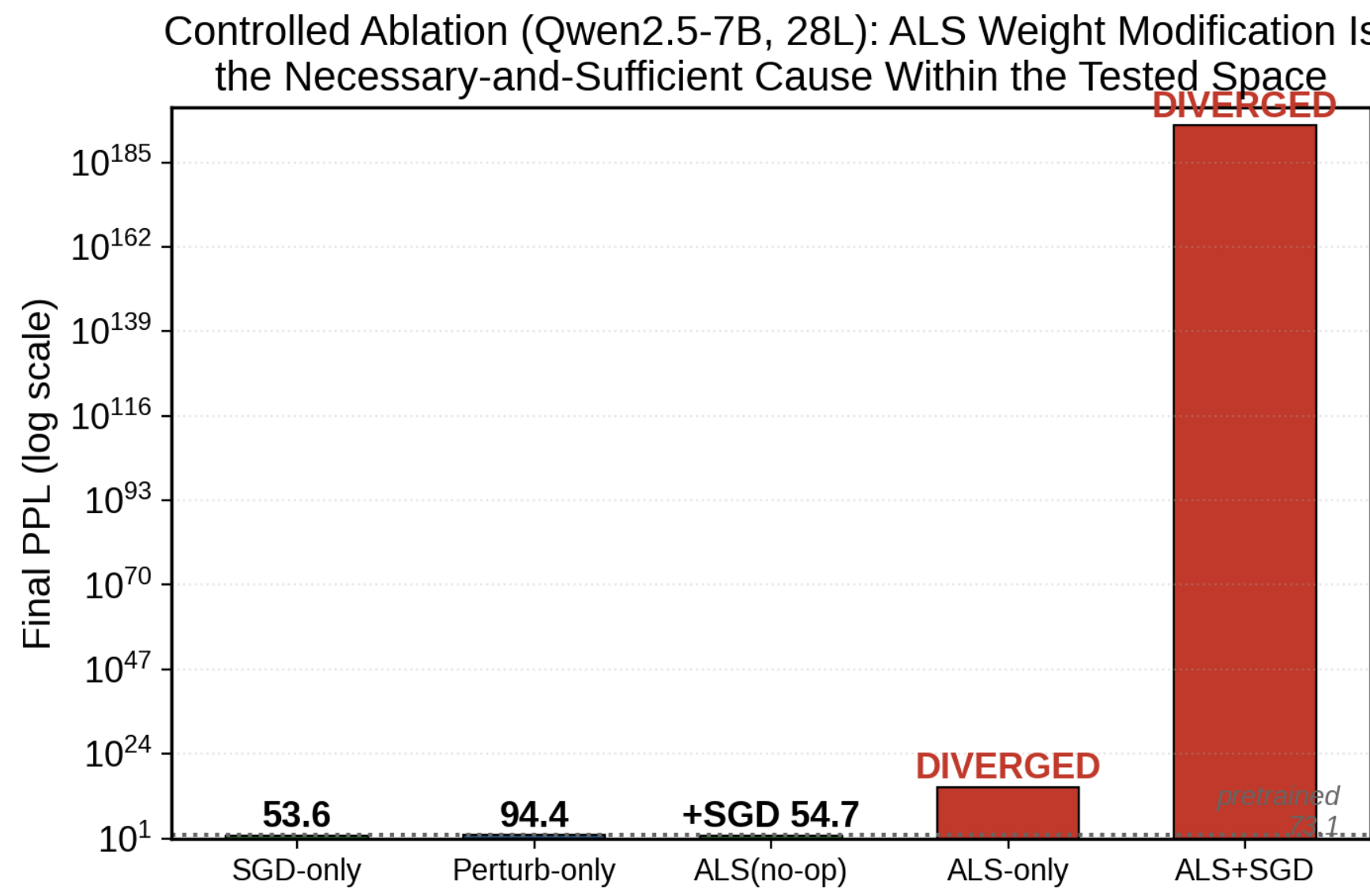


The theory predicts the depth boundary **between 25 and 28 layers**, consistent with the observed 24-converges / 28-diverges split across 8 architectures. The perturbation *grows* across cycles (δ : 0.085 → 0.196) — a positive feedback loop, which is why damping/clipping patches only delay the failure.

Key Findings & Experimental Results

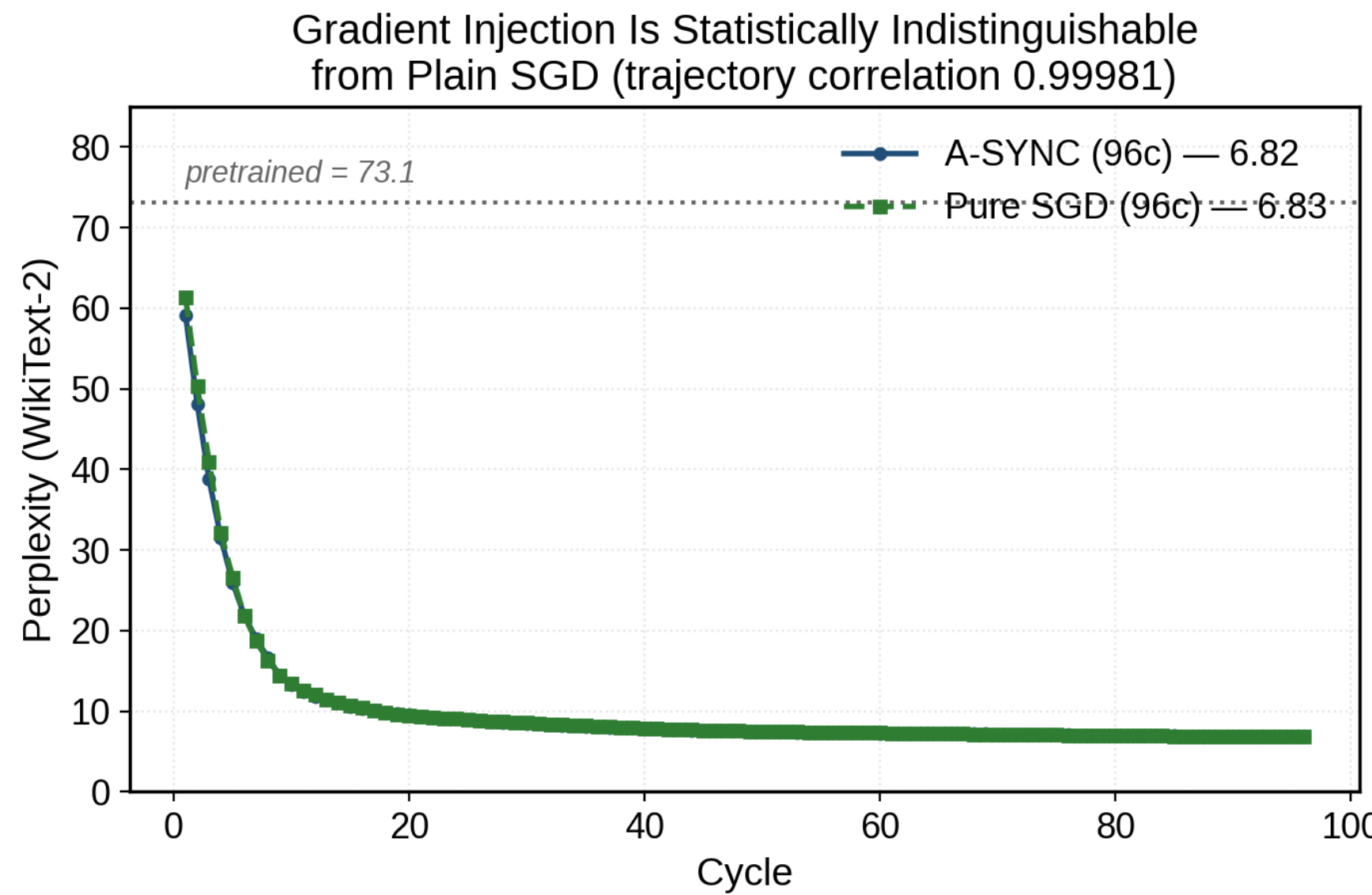
Finding 1 — ALS weight modification is the sole cause of divergence

A five-condition ablation on Qwen2.5-7B isolates the cause: every condition with a real weight solve diverges; SGD-only, perturbation-only and suppressed-weight ALS all converge on the same model.



Finding 2 — Gradient injection: statistically indistinguishable from plain SGD

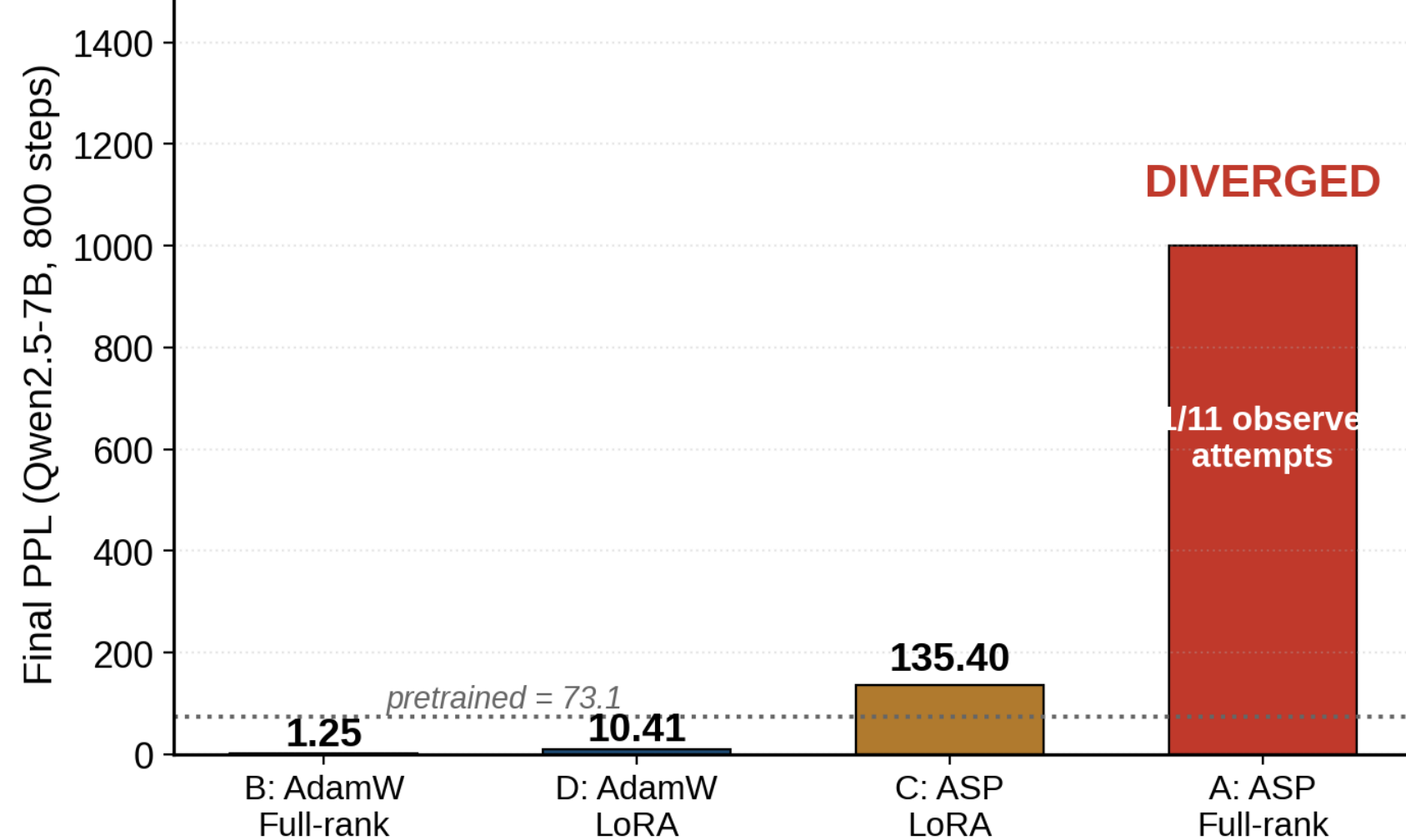
A-SYNC injects the ALS solution as a gradient bias; matched-budget pure SGD reproduces its trajectory (correlation 0.99981, mean per-cycle diff 0.116 PPL). No tested schedule beats plain SGD.



Finding 3 — The 2×2 matrix: AdamW below baseline, ASP diverges

On Qwen2.5-7B (800 steps, N=3) vs the 73.1 pretrained baseline: AdamW full-rank 1.25 PPL, AdamW+LoRA 10.41, ASP+LoRA 135.4, ASP full-rank diverges.

2×2 Factorial: AdamW Below Baseline; ASP Diverges Under Full-Rank



References & Reproducibility

- [1] Hu, E. et al. LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022.
- [2] Loshchilov, I., Hutter, F. Decoupled Weight Decay Regularization. ICLR 2019.
- [3] Koren, Y. et al. Matrix Factorization Techniques for Recommender Systems. IEEE Computer 42(8), 2009.
- [4] Merity, S. et al. Pointer Sentinel Mixture Models. ICLR 2017.
- [5] Hinton, G. et al. Distilling the Knowledge in a Neural Network. arXiv:1503.02531.

All experiments reproducible: runs/ + experiments/ in the project repo. Evaluation-harness audit and per-step data archived. Scan to view the repository:



Takeaway: on deep models, remove the ALS machinery and let gradient descent do its job with a sustained learning rate — AdamW 58× better than the pretrained baseline, plain SGD 10.7×, LoRA 7×. Best quality still requires AdamW.

Acknowledgements: supervised by Prof. Hoi To Wai (CUHK CSE). All experiments run on an RTX 5090; data, seeds, and per-step logs archived in the repository. Results verified with a corrected evaluation harness (tokenizer-matched, pad-masked labels).